COMP9602: Optimization

Duality (II)

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Next

- Interpretations of duality
  - Geometric interpretation
  - Sensitivity interpretation
  - Saddle-point interpretation
  - Game interpretation

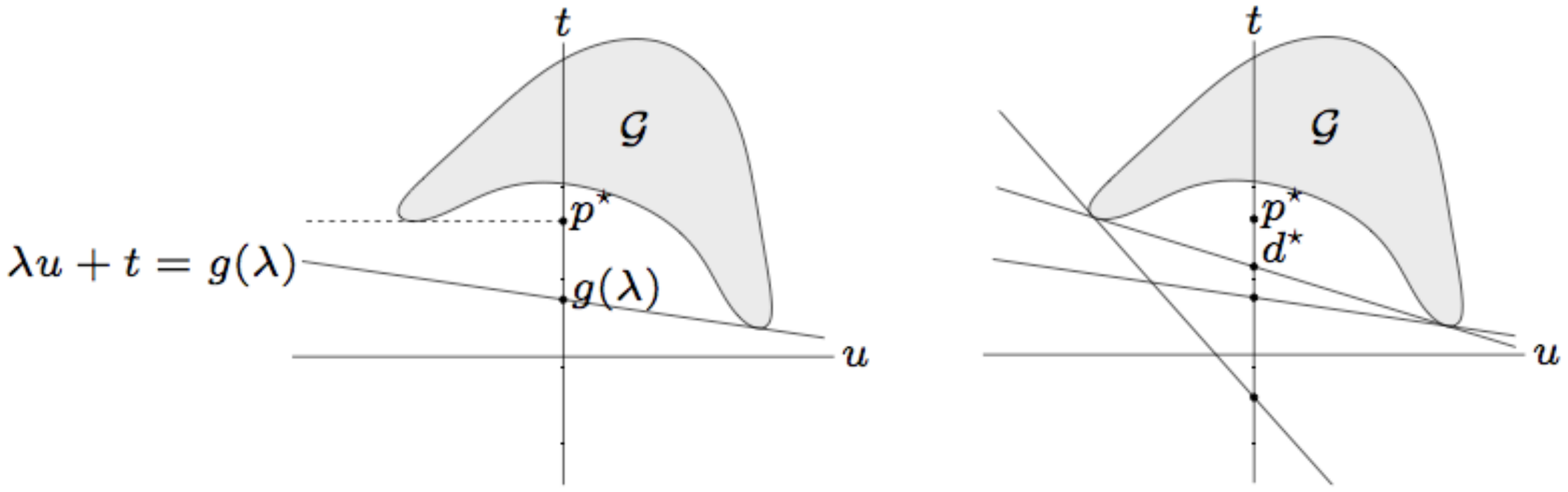
- Examples
  - usage of duality
  - non-convex problem with strong duality

Geometric interpretation

for simplicity, consider problem with one constraint $f_1(x) \leq 0$

interpretation of dual function:

$$g(\lambda) = \inf_{(u,t)\in\mathcal{G}} (t + \lambda u), \qquad \text{where} \quad \mathcal{G} = \{(f_1(x), f_0(x)) \mid x \in \mathcal{D}\}$$
$$u = f_1(x), t = f_0(x)$$

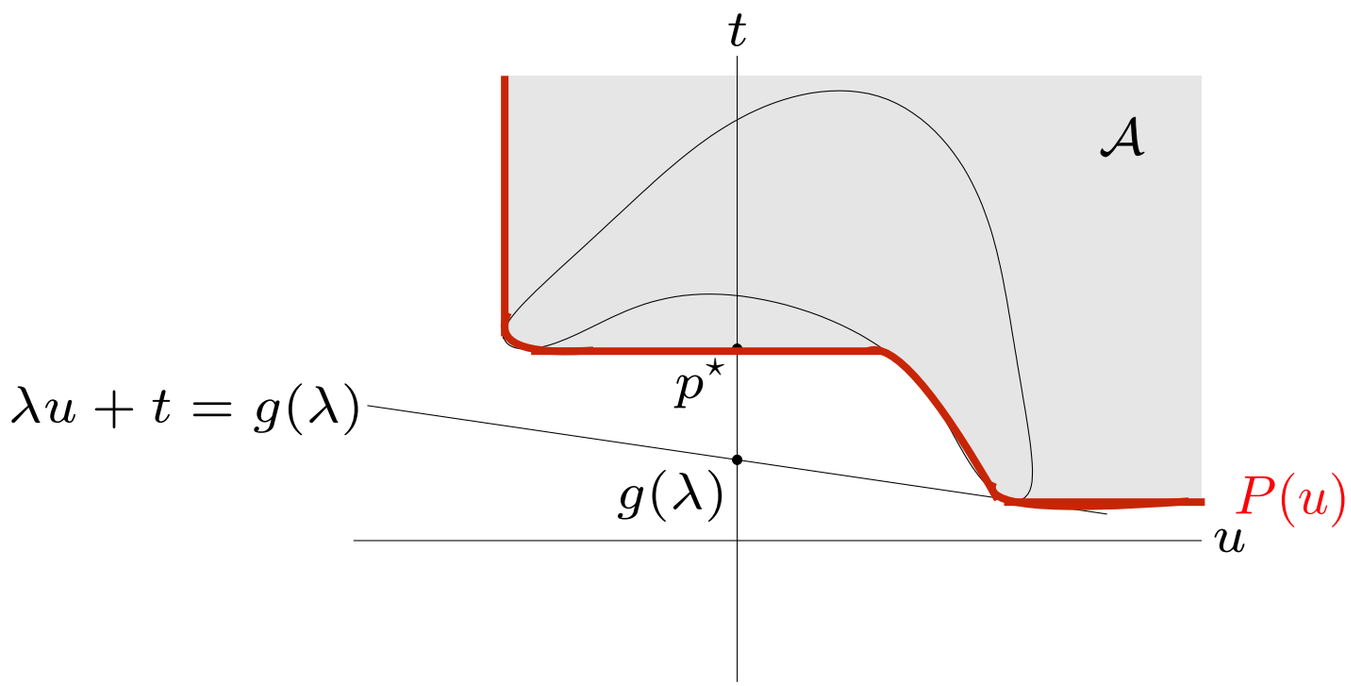


- $\lambda u + t = g(\lambda)$ is (non-vertical) supporting hyperplane to $\mathcal{G}$
  which intersects t-axis at $t = g(\lambda)$

Geometric interpretation (cont'd)

epigraph variation: same interpretation if $\mathcal{G}$ is replaced with

$$\mathcal{A} = \{(u,t) \mid f_1(x) \le u, f_0(x) \le t \text{ for some } x \in \mathcal{D}\}$$



$$P(u) = \min f_0(x) \quad \text{s.t. } f_1(x) \le u$$

strong duality

- holds if there is a non-vertical supporting hyperplane to $\mathcal{A}$ at $(0, p^\star)$
- for convex problem, $\mathcal{A}$ is convex, hence has supp. hyperplane at $(0, p^\star)$ P(u) is convex
- Slater's condition: if there exist $(\tilde{u}, \tilde{t}) \in \mathcal{A}$ with $\tilde{u} < 0$, then supporting hyperplanes at $(0, p^\star)$ must be non-vertical

Proof of P(u) being convex for convex primal problem:

Consider $P(u_1) = \min_{\text{s.t. } f_1(x) \le u_1} f_0(x)$ with optimal point x_1^*

$P(u_2) = \min_{\text{s.t. } f_1(x) \le u_2} f_0(x)$ with optimal point x_2^*

$P(\alpha_1 u_1 + \alpha_2 u_2) = \min_{\text{s.t. } f_1(x) \le \alpha_1 u_1 + \alpha_2 u_2} f_0(x)$

We have $f_1(\alpha_1 x_1^* + \alpha_2 x_2^*) \le \alpha_1 f_1(x_1^*) + \alpha_2 f_1(x_2^*)$ (because f1(x) is convex)

$\le \alpha_1 u_1 + \alpha_2 u_2$

Then $f_0(\alpha_1 x_1^* + \alpha_2 x_2^*) \ge \min_{\text{s.t. } f_1(x) \le \alpha_1 u_1 + \alpha_2 u_2} f_0(x)$

$$\alpha_1 P(u_1) + \alpha_2 P(u_2) = \alpha_1 f_0(x_1^*) + \alpha_2 f_0(x_2^*)$$

$$\ge f_0(\alpha_1 x_1^* + \alpha_2 x_2^*) \text{ (because f0(x) is convex)}$$

$$\ge P(\alpha_1 u_1 + \alpha_2 u_2)$$

Therefore, P(u) is convex

Sensitivity interpretation

(unperturbed) optimization problem and its dual

$$\begin{array}{lll} \text{minimize} & f_0(x) & \\ \text{subject to} & f_i(x) \le 0, & i = 1, \ldots, m \\ & h_i(x) = 0, & i = 1, \ldots, p \end{array} \qquad \begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \succeq 0 \end{array}$$

perturbed problem and its dual

$$\begin{array}{lll} \text{min.} & f_0(x) & \\ \text{s.t.} & f_i(x) \le u_i, & i = 1, \ldots, m \\ & h_i(x) = v_i, & i = 1, \ldots, p \end{array} \qquad \begin{array}{ll} \text{max.} & g(\lambda, \nu) - u^T\lambda - v^T\nu \\ \text{s.t.} & \lambda \succeq 0 \end{array}$$

- $p^\star(u, v)$ is optimal value as a function of u, v

- **Local sensitivity**: if strong duality holds and p*(u,v) is differentiable at (0,0)

$$\lambda_i^\star = -\frac{\partial p^\star(0,0)}{\partial u_i}, \qquad \nu_i^\star = -\frac{\partial p^\star(0,0)}{\partial v_i}$$

Saddle-point interpretation

- Assume no equality constraints (results can be easily extended)

$$p^* = \inf_x \sup_{\lambda \succeq 0} L(x, \lambda) \qquad\qquad d^* = \sup_{\lambda \succeq 0} \inf_x L(x, \lambda)$$

- Weak duality: $\sup_{\lambda \succeq 0} \inf_x L(x, \lambda) \leq \inf_x \sup_{\lambda \succeq 0} L(x, \lambda)$

- Strong duality: $\sup \inf L(x, \lambda) = \inf \sup L(x, \lambda)$

- Max-min inequality generally holds:

$$\sup_{z \in Z} \inf_{w \in W} f(w, z) \leq \inf_{w \in W} \sup_{z \in Z} f(w, z) \qquad \text{for any } f, W, Z$$

- Strong max-min property (or saddle-point property) holds if

$$\sup_{z \in Z} \inf_{w \in W} f(w, z) = \inf_{w \in W} \sup_{z \in Z} f(w, z)$$

Saddle-point interpretation (cont'd)

- Saddle-point for f: a pair $\tilde{w} \in W, \tilde{z} \in Z$ that satisfy

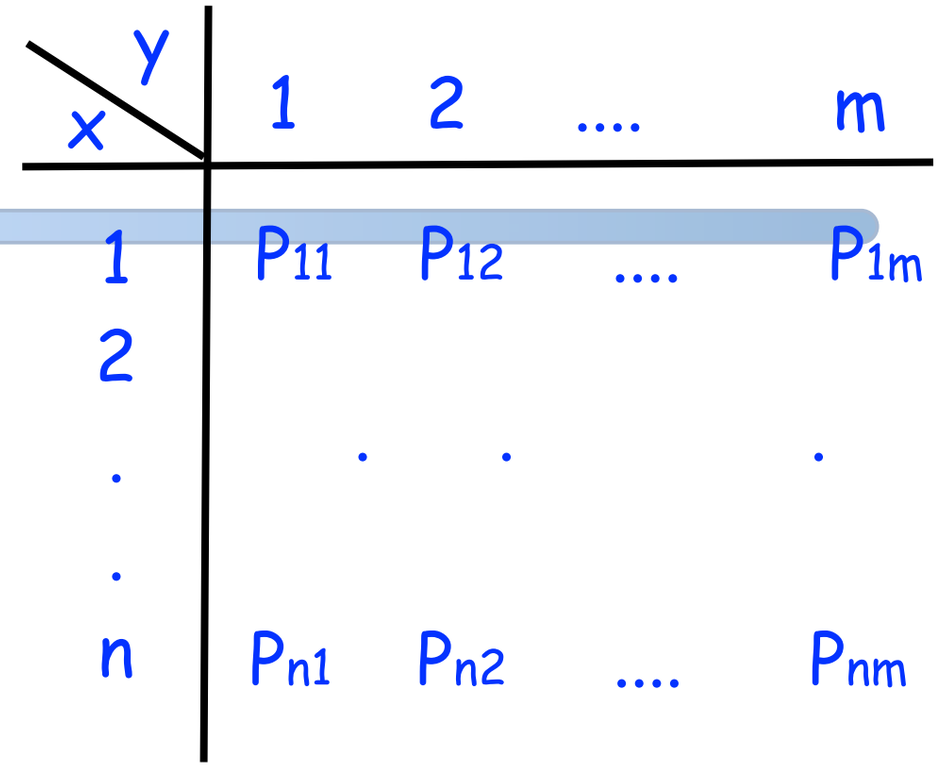
$$f(\tilde{w}, z) \leq f(\tilde{w}, \tilde{z}) \leq f(w, \tilde{z}), \forall w \in W, z \in Z$$

<=> $f(\tilde{w}, \tilde{z}) = \inf_{w \in W} f(w, \tilde{z})$ $\qquad$ $f(\tilde{w}, \tilde{z}) = \sup_{z \in Z} f(\tilde{w}, z)$

<=> strong max-min property holds

- x^*, λ^* are primal and dual optimal points and strong duality holds
  <=> (x^*, λ^*) form a saddle-point for the Lagrangian

Game interpretation



two-player zero-sum game

- Player x chooses strategy from 1, 2, ..., n
- Player y chooses strategy from 1, 2, ..., m
- Pij is the amount x pays to y (payoff) when x plays strategy i, y plays strategy j

- mixed strategy:

  ui: prob(player x chooses strategy i)

  vj: prob(player y chooses strategy j)

- expected payoff: $u^T P v$

Game interpretation (cont'd)

- Suppose x fixes strategy u, then y plays (decides v) to maximize expected payoff:

$$\max u^T P v$$
$$\text{s.t.} \quad \sum_{i=1}^{m} v_i = 1$$
$$v \succeq 0$$

=> optimal value: $\max_{i=1,\ldots,m} (P^T u)_i$ (ith row)

=> optimal point: $v_j = 1, j = \text{argmax}_i (P^T u)_i$
$v_i = 0, \forall i \neq j$

- So x must choose u to minimize $\max_{i=1,\ldots,m} (P^T u)_i$:

$$\min_u \max_{i=1,\ldots,m} (P^T u)_i$$
$$\text{s.t.} \quad \sum_{i=1}^{n} u_i = 1$$
$$u \succeq 0$$

=>

$$\min t$$
$$\text{s.t.} \quad P^T u \preceq t\mathbf{1}$$
$$u^T \mathbf{1} = 1$$
$$u \succeq 0$$

LP (1) with Optimal value: p_1^*

Game interpretation (cont'd)

- Suppose y plays first (v given), then x chooses u to minimize expected payoff:

$$\min u^T P v$$

s.t. $\sum_{i=1}^{n} u_i = 1$

$$u \succeq 0$$

=> optimal value: $\min_{i=1,\ldots,n} (Pv)_i$ (ith row)

=> optimal point: $u_j = 1, j = \text{argmin}_i (Pv)_i$

$u_i = 0, \forall i \neq j$

- So y chooses v to maximize $\min_{i=1,\ldots,n} (Pv)_i$:

$$\max_{v} \min_{i=1,\ldots,n} (Pv)_i$$

s.t. $\sum_{i=1}^{m} v_i = 1$

$$v \succeq 0$$

=>

$$\max t$$

s.t. $Pv \succeq t\mathbf{1}$

$v^T \mathbf{1} = 1$

$v \succeq 0$

LP (2) with Optimal value: p_2^*

Game interpretation (cont'd)

LP (1) and LP (2) are duals of each other and thus have the same optimal values: $p_1^* = p_2^*$

Therefore, there is no advantage to play second, i.e., $p_1^* \not\geq p_2^*$

Game interpretation (cont'd)

- Consider payoff function $f(u,v) = u^T P v$, the optimum u* for LP (1) and the optimum v* for LP (2) form a saddle-point for f(u,v)
  - f(u*,v) ≤ f(u*,v*) ≤ f(u,v*)

<=> $f(u^*, v^*) = \inf_u f(u, v^*)$ $f(u^*, v^*) = \sup_v f(u^*, v)$

- Nash equilibrium of the game: (u*,v*) such that

  u* is the best response of player x with respect to v*

  v* is the best response of player y with respect to u*

Usage example of duality I

- Duality gives a way to analytically solve an optimization problem
  - example:

$$\min \| x \|_2^2$$
$$\text{s.t.} \quad Ax = b$$

Lagrangian: $L(x, \nu) = x^T x + \nu^T (Ax - b)$

Lagrange dual function: $g(\nu) = \min x^T x + \nu^T Ax - \nu^T b$

$$\frac{\partial(x^T x + \nu^T Ax)}{\partial x} = 2x^T + \nu^T A = 0 \quad \Rightarrow \quad x = -\frac{A^T \nu}{2}$$

$$g(\nu) = -\frac{1}{4}\nu^T AA^T \nu - \nu^T b$$

Dual problem: $\max_{\nu} -\frac{1}{4}\nu^T AA^T \nu - \nu^T b$

$$\frac{\partial(-\frac{1}{4}\nu^T AA^T \nu - \nu^T b)}{\partial \nu} = 0 \quad \Rightarrow \quad \nu = -2(AA^T)^{-1} b$$

$$x = A^T (AA^T)^{-1} b$$

Usage example of duality II

- Duality is useful to find non-trivial lower bounds to non-convex minimization problems
  - example:

$$\min x^T W x \qquad W \succeq 0$$

s.t.

$$x_i \in \{-1, 1\}, \forall i = 1, \ldots, n$$

  - relaxation:

$$\min x^T W x$$

s.t.

$$-1 \leq x_i \leq 1, i = 1, \ldots, n$$

solution to this problem: 0, which is a trivial lower bound

  - use duality to obtain better lower bound (as on page 15 of 7_Duality_I_COMP9602_Fall2026.pdf)

Non-convex problem with strong duality

- Strong duality ~~=>~~ convex problem
  - example non-convex problem with strong duality:

$$\min x^T A x \qquad A \in S^n$$

$$\text{s.t.} \quad x^T x = 1$$

solution: $\min_{||x||_2=1} x^T A x = \text{minimum eigenvalue of A}$

let $x = \sum_{i=1}^{n} \alpha_i v_i$ (linear combination of orthogonal eigenvectors of A)

v_i is an eigenvector corresponding to eigenvalue λ_i : $Av_i = \lambda_i v_i$, $||v_i||_2 = 1$

we have $x^T A x = (\sum_{i=1}^{n} \alpha_i v_i)^T A (\sum_{i=1}^{n} \alpha_i v_i) = \sum_{i=1}^{n} \alpha_i^2 \lambda_i ||v_i||_2^2 = \sum_{i=1}^{n} \alpha_i^2 \lambda_i$

since $x^T x = 1$, we know $\sum_{i=1}^{n} \alpha_i^2 = 1$

therefore, $x^T A x$ is minimised when $\alpha_j = 1$ for $\lambda_j = \min\{\lambda_1, \ldots, \lambda_n\}$, $\alpha_i = 0$ for $i \neq j$

and $x^* = v_j$

Non-convex problem with strong duality (cont'd)

- Strong duality ~~=>~~ convex problem
  - example non-convex problem with strong duality:

$$\min x^T A x \qquad A \in S^n$$
$$\text{s.t.} \quad x^T x = 1$$

Derive the dual problem

Lagrangian: $L(x, \nu) = x^T A x + \nu(x^T x - 1)$

Lagrange dual function: $g(\nu) = \min_x L(x, \nu) = \min_x x^T (A + \nu I) x - \nu$

$$= \begin{cases} -\nu, \text{if } A + \nu I \succeq 0, \\ -\infty, \text{ otherwise.} \end{cases}$$

Dual problem:

$$\max -\nu$$
$$\text{s.t. } A + \nu I \succeq 0$$

$$A + \nu I = Q^T \begin{bmatrix} \lambda_1 & & \\ & \dots & \\ & & \lambda_n \end{bmatrix} Q + Q^T \begin{bmatrix} \nu & & \\ & \dots & \\ & & \nu \end{bmatrix} Q = Q^T \begin{bmatrix} \lambda_1 + \nu & & \\ & \dots & \\ & & \lambda_n + \nu \end{bmatrix} Q$$

Therefore, the largest $-\nu$ to make $A + \nu I \succeq 0$ is $\min\{\lambda_1, \dots, \lambda_n\}$

$d^* =$ minimum eigenvalue of A

- Reference
  - Chapter 5.3 - 5.4, 5.6, Convex Optimization.
- Acknowledgement
  - Some materials are extracted from the slides of Stanford EE364a and EE364b
  - Some materials are extracted from the lecture notes of Convex Optimization by Prof. Wei Yu at the University of Toronto